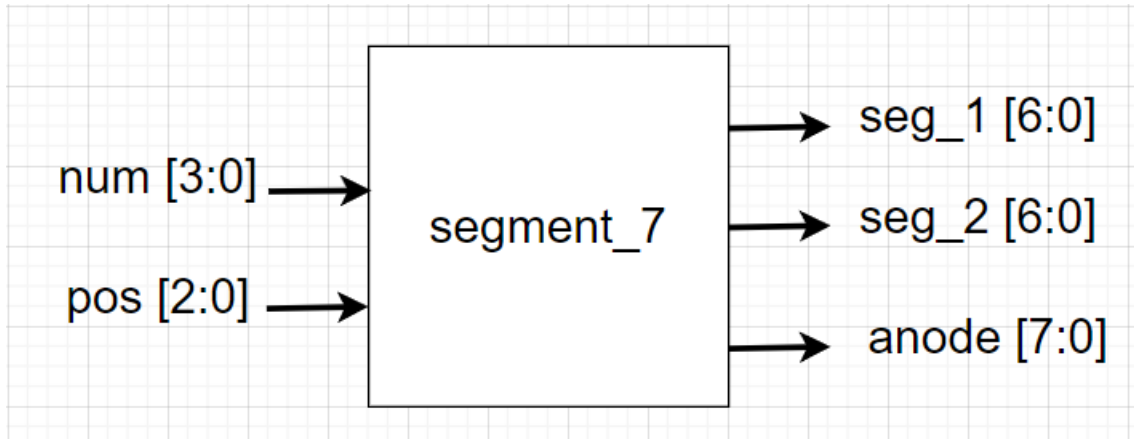


Digital Logic Design – Lab 4

1. **Topic:** 7 Segment Display
2. **Design structure:**



3. **Code:** the code is also provided along with this report on the same .zip

```
23 module segment_7(num, pos, anode, seg_1, seg_2);  
24     // input ports -> correspond to the switches  
25     input [3:0] num;  
26     input [2:0] pos;  
27  
28     // output ports -> corresponds to which LED will be displayed,  
29     // and which number will be shown  
30     output [7:0] anode;  
31     output [6:0] seg_1, seg_2;  
32  
33     reg [7:0] anode;  
34     reg [6:0] seg_1, seg_2;
```

```

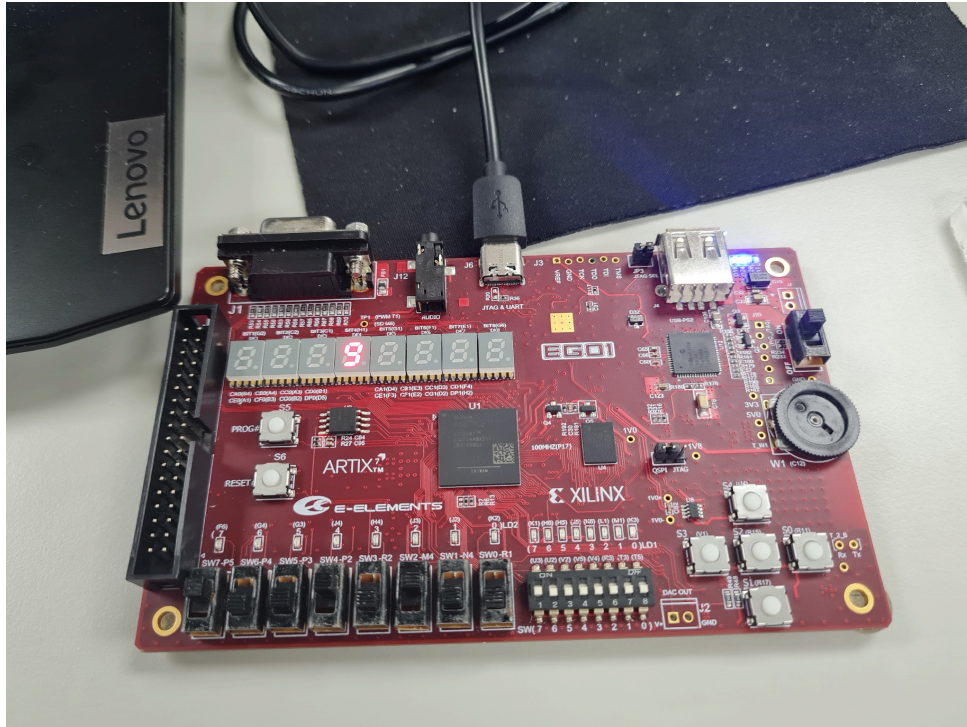
36 // For the 7 bit binary number, 1 means light up LED
37 // This always block displays the number on the LED
38 always @(num)
39     case (num)
40         4'b0000: {seg_1, seg_2} = {7'b1111110, 7'b1111110}; // "0"
41         4'b0001: {seg_1, seg_2} = {7'b0110000, 7'b0110000}; // "1"
42         4'b0010: {seg_1, seg_2} = {7'b1101101, 7'b1101101}; // "2"
43         4'b0011: {seg_1, seg_2} = {7'b1111001, 7'b1111001}; // "3"
44         4'b0100: {seg_1, seg_2} = {7'b0110011, 7'b0110011}; // "4"
45         4'b0101: {seg_1, seg_2} = {7'b1011011, 7'b1011011}; // "5"
46         4'b0110: {seg_1, seg_2} = {7'b1011111, 7'b1011111}; // "6"
47         4'b0111: {seg_1, seg_2} = {7'b1110000, 7'b1110000}; // "7"
48         4'b1000: {seg_1, seg_2} = {7'b1111111, 7'b1111111}; // "8"
49         4'b1001: {seg_1, seg_2} = {7'b1111011, 7'b1111011}; // "9"
50         4'b1010: {seg_1, seg_2} = {7'b1110111, 7'b1110111}; // A
51         4'b1011: {seg_1, seg_2} = {7'b0011111, 7'b0011111}; // B
52         4'b1100: {seg_1, seg_2} = {7'b1001110, 7'b1001110}; // C
53         4'b1101: {seg_1, seg_2} = {7'b0111101, 7'b0111101}; // D
54         4'b1110: {seg_1, seg_2} = {7'b1001111, 7'b1001111}; // E
55         default: {seg_1, seg_2} = {7'b1000111, 7'b1000111}; // F
56     endcase

58 // This always block controls which LED will light up
59 always @(pos)
60     case (pos)
61         3'b000: anode = 8'b00000001;
62         3'b001: anode = 8'b00000010;
63         3'b010: anode = 8'b00000100;
64         3'b011: anode = 8'b00001000;
65         3'b100: anode = 8'b00010000;
66         3'b101: anode = 8'b00100000;
67         3'b110: anode = 8'b01000000;
68         default: anode = 8'b10000000;
69     endcase
70 endmodule

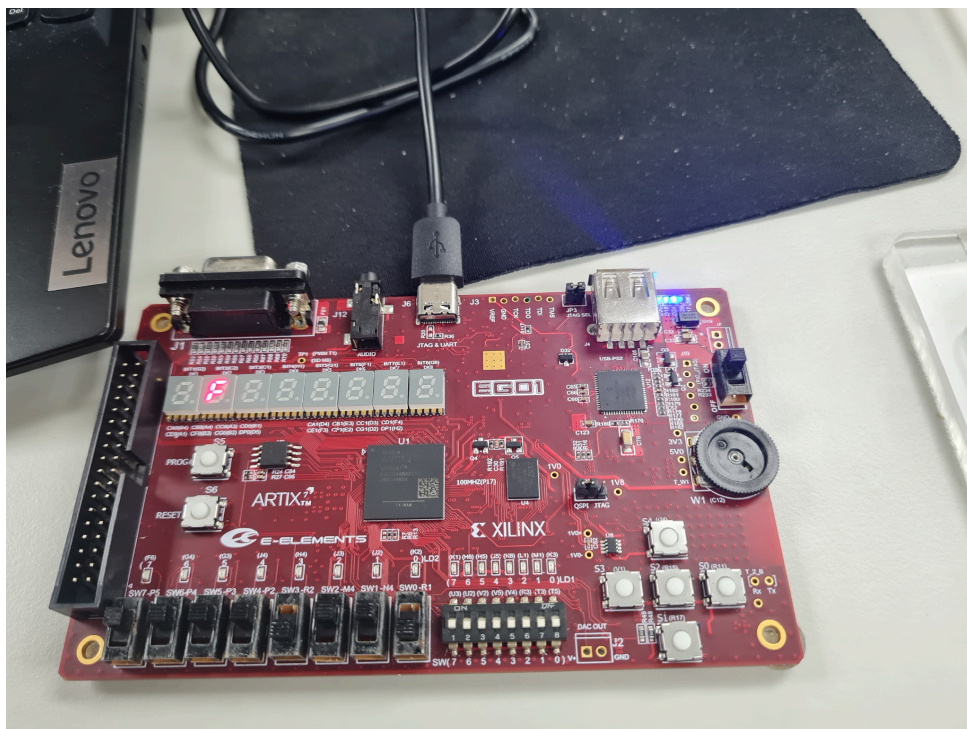
```

4. **Result:** the demo of the working FPGA board can be seen in the following link. First, I show how the 3 right-most switches control the position of the LED being lighted, then I show that the 4 left-most switches control the number being displayed

<https://www.youtube.com/watch?v=zdD3uvG7wCA>



Right-most switches are (100) so LED at position 4 is displayed. Left-most switches are (1001) so the number displayed is 9



Right-most switches are (110) so LED at position 6 is displayed. Left-most switches are (1111) so the number displayed is F

5. Reflection

In this lab, we displayed the 7 segments LED display of a FPGA board. This was a good first experience deploying our code into a board, and by doing so, learning how LEDs are being displayed in pretty much every other device you encounter in your day-to-day life